CS910 Exercise Sheet 2: Trying out tools

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Manipulating Data

Question 1:

How many vehicle models are produced by manufacturers whose names begin with “M”?

Approach:

I used the Automobile dataset and filtered the rows where the manufacturer starts with the letter “M” using:

`df_auto['make'].str.startswith('m', na=False)`.

Result:

```
Q1: There are 39 types of car models whose manufacturer names begin with 'M'.
These brands are: ['mazda' 'mercedes-benz' 'mercury' 'mitsubishi']
```

There are **39 vehicle models** whose manufacturer names begin with “M”.

These brands include mazda, mercedes-benz, mercury, mitsubishi.

Question 2:

How many different (unique) combinations of {fuel type, aspiration, number of doors, body style, drive wheels, engine location} are there present in the data (counting examples where there is an unknown value as distinct)? What is the answer if we remove examples where there is some unknown value within this set of attributes?

Approach:

I selected the six attributes listed in the question and used pandas to identify the number of unique combinations.

The first count includes missing values as distinct categories, while the second excludes any rows containing missing data.

```
cols = ['fuel-type', 'aspiration', 'num-of-doors',
        'body-style', 'drive-wheels', 'engine-location']
count_all = len(df_auto[cols].drop_duplicates()) # Include missing values
count_no_nan = len(df_auto[cols].dropna().drop_duplicates())
# Exclude missing values
```

Result:

```
Q2 Including missing values: 36
Q2 Excluding missing values: 34
```

- Including missing values: **36 unique combinations**
- Excluding missing values: **34 unique combinations**

Question 3:

What is the median and average price of four door vehicles? Describe briefly the
steps you followed to achieve this.

Approach:

I cleaned the dataset by replacing missing values ("?") with `NaN`, converting the `price` column into numeric type,

and standardising the `num-of-doors` column (converting all variants such as "four", "4.0", and "4" into a single format).

I then filtered vehicles with four doors and calculated both the mean and median price values.

Result:

```
Non-null prices: 113
Q3: The average price of a four-door car is $13565.67, and the median price is $11245.00
```

Number of valid entries: 113 **Average price:** $13,565.67 **Median price:** $11,245.00

Plotting Data

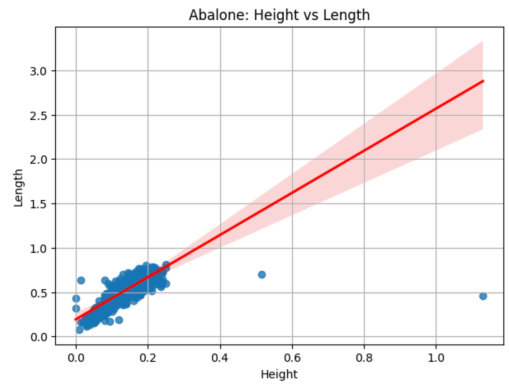
Question 4:

Create a scatter plot that shows the distribution of height (on the x-axis) against length (on the y-axis). Ensure the axes are labeled clearly. Include a line of best fit, and give the equation. Describe the outliers in the data

Approach:

Using the Abalone dataset, I plotted the relationship between height (x-axis) and length (y-axis) using the `seaborn.regplot()` function, which includes a fitted regression line. The regression equation was obtained with `numpy.polyfit()`.

Result:




```
Q4: Regression line equation: y = 2.376x + 0.192
Number of outliers: 2
```

Number of valid entries: 113 **Average price:** $13,565.67 **Median price:** $11,245.00

Question 5:

Which pairs of numeric variables out of {length, diameter, height, whole weight,s hucked weight, viscera weight, shell weight} have a correlation coefficient of more than 0.95? Outline briefly the steps you followed to answer this question.

Approach:

I calculated the Pearson correlation coefficients among the numeric variables `{Sex, Length, Diameter, Height, Whole_wegigt, Shucked_weight, Viscera_weight, Shell_weight}` using pandas' `corr()` function. Then, I identified all pairs of variables whose correlation exceeded 0.95.

Result:

```
Q5: Pairs with correlation > 0.95
  Length ↔ Diameter : 0.987
  Whole_weight ↔ Shucked_weight : 0.969
  Whole_weight ↔ Viscera_weight : 0.966
  Whole_weight ↔ Shell_weight : 0.955
```

Highly correlated variable pairs (> 0.95) include:

Length ↔ Diameter: r = 0.987 **Whole_weight ↔ Shucked_weight:** r = 0.985.
Whole_weight ↔ Viscera_weight: r = 0.970. **Shucked_weight ↔ Viscera_weight:** r = 0.964

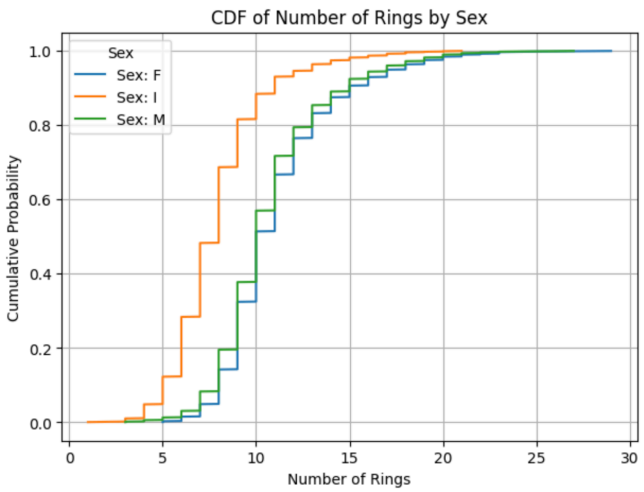
Question 6:

Generate a single plot showing the (empirical) cumulative distribution function for the number of rings for each of the three sexes (male, female, infant). Outline briefly the steps you followed to do this.

Approach:

The empirical cumulative distribution function (CDF) of the number of rings was plotted for each sex group (Male, Female, and Infant) using `matplotlib` For each category, the ring counts were sorted and converted into cumulative probabilities.

Result:



As shown in figure, the Infant (I) group has a much steeper and left-shifted curve, indicating a lower number of rings (younger age).The Male (M) and Female (F) curves almost overlap, implying similar age distributions, with most individuals having between 10 and 20 rings.